

Math 421

Wednesday, September 24

Announcements

- Homework 3 due Friday
- Grace's Drop-In Hours
 - **Today:** 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4

Activity & Discussion Question

Find the supremum/infimum of the following sets (if they exist). ❖ Is the empty set bounded?

1. (a, b)

2. $\{n \in \mathbb{N} : n^2 < 3\}$

3. $\{x \in \mathbb{R} : x^2 < 3\}$

4. $\left\{\sin \frac{n\pi}{4} : n \in \mathbb{N}\right\}$

Completeness Axiom

Completeness Axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound. In other words, $\sup S$ exists and is a real number.

Q: Does the completeness axiom hold for $\mathbb{Q}$?

Corollary: Every nonempty subset S of $\mathbb{R}$ that is bounded below has a greatest lower bound.

Unbounded Sets & Notation

Q: What does it mean for a set to be **unbounded**?

Equivalent Definition of $\sup S$

Thm. Let $M \in \mathbb{R}$ be an upper bound for a set $S \subseteq \mathbb{R}$. Then $M = \sup S$ if and only if for every $\varepsilon > 0$ there exists some $s \in S$ with $M - \varepsilon < s$.

Discussion Questions:

- ❖ Describe what this theorem is saying in your own words.
- ❖ Convince yourselves that this theorem is true, it might be helpful to think in terms of a specific example and then see how it can generalize.

Archimedean Property

Thm. Let $a, b \in \mathbb{R}$. If $a > 0$ and $b > 0$, then for some positive integer n , we have $na > b$.

Discussion Questions:

- ❖ What is this theorem saying in your own words?
- ❖ What does this property say if $a = 1$?
- ❖ What does this property say if $b = 1$?

Archimedean Property

Thm. Let $a, b \in \mathbb{R}$. If $a > 0$ and $b > 0$, then for some positive integer n , we have $a > \frac{b}{n}$.

Discussion.

Archimedean Property

Proof.

Density of $\mathbb{Q}$

Thm. If $a, b \in \mathbb{R}$ and $a < b$, then there is a rational $r \in \mathbb{Q}$ such that $a < r < b$.

Proof Sketch.

Recipe to Show $\sup S = M$

Example 1: Proof Sketch Using Recipe

Theorem. Assume $A \subseteq [0, \infty)$ is nonempty and bounded above. If $B = \{a^2 : a \in A\}$, then $\sup B = (\sup A)^2$.

Proof Sketch.

Example 2: Proof Sketch Using Recipe

Theorem. Let $A = [a, b)$. Then $\sup A = b$.

Proof Sketch.